Olist SQL Analysis

One-Page Insight Summary | Cristine Angelica Marie Serrano Tolentino

Data Overview

I looked at the Olist online marketplace data in MySQL, using six tables that cover orders, items, customers, payments, reviews and sellers. There are 96,478 delivered orders worth R$15,422,462, placed between October 2016 and August 2018. Two things I had to be careful about: the customers table gives a new customer_id for every order, so I had to use customer_unique_id to find the real person, and the items table counts items while the payments table counts payments, so joining them together would double up my money. The first few months are also incomplete, since November 2016 has no delivered orders and December 2016 has only one order worth R$19.62.

Three Business Insights

1. **The growth came from more customers, not bigger orders, and it has stopped.** Sales went from R$127,546 in January 2017 up to R$1,153,528 in November 2017, which is nine times bigger in ten months (Q2). The number of orders grew the same way, from 750 to 7,289. But the average amount people spent per order never changed, it just stayed between R$146 and R$170 the whole time. So all of that growth came from more people buying, not from people spending more. Then Q3 shows what happened after. In 2017 the monthly growth was between +45% and +112%. From February 2018 the numbers moved between -10% and +16%, and the ups and downs mostly cancelled each other out. No month ever beat November 2017 again.

2. **Almost nobody comes back, and that is why the growth could not last.** Q7 shows 90,557 people who bought only once, against 2,801 who looked like they came back. But when I checked the gap between their orders, 890 of those second orders happened on the same day as the first, sometimes only seconds apart. Those are not people coming back. That is one shopping basket getting split into separate orders because the items came from different sellers. The real number is 2,015 customers, which is 2.2%, and they took an average of 110.7 days to come back. This is exactly why the growth stopped. With almost nobody returning, the business has to find a whole new set of buyers every month, and then even more on top of that, just to grow. That only works while it keeps finding more and more new people.

3. **No single category or customer group carries the business, and the tiers are not what they look like.** The biggest category, health and beauty, is only 9.33% of product sales (Q8). The customer tiers are spread out too, at 42.7% Medium, 31.8% Low and 25.5% High (Q6). Nothing here is big enough to build a plan around. And there is something else. Since 97% of customers only buy once, their total spend is really just their first order. So these tiers are not loyalty groups at all, they are order size groups. A High Value customer spending R$923 is not a loyal customer, it is someone who bought one expensive thing and left. Q1 proves it, because the highest spending customer in the whole dataset placed only one order.

Two Recommendations

1. **Make the first order bigger instead of building a loyalty program.** Since most people only buy once, their first order is basically all the money we will ever get from them. So the goal should be to make that one order bigger. Add product bundles, suggested add-ons at checkout, and free shipping once the customer spends a bit more than the R$159.86 average. A points card would not help much because there are hardly any repeat customers to give points to.

2. **Reach out to customers 60 to 75 days after they buy.** The few customers who do come back take about 110.7 days to do it. So we should reach out before that, while they still remember us, instead of waiting until they are gone. Start with the Low Value group, because at 61,804 customers spending R$79.29 each it is by far the biggest group, so even a small improvement there is worth more than a big one at the top.